

- (b) State the mass and the charge, in terms of the atomic mass unit  $u$  and the elementary charge  $e$ , of an alpha particle.

mass = .....  $u$

charge = .....  $e$   
[2]

- (c) There are two types of hadron.  
The hadrons that are in alpha particles are each composed of three quarks.

- (i) State the name of this type of hadron.

..... [1]

- (ii) Show, by reference to their constituent quarks, that the hadrons in an alpha particle have charges of either zero or  $+1e$ .

[2]

[Total: 10]

